AP-001 Ontology-driven GraphRAG with grounded answers

The architecture pattern: ontology, extraction with provenance, the graph as record, grounded answering

Reversed from the implementation at 0.3.0. Draft.

Context

An organisation has documents and wants to answer questions from them with every statement checkable. Retrieval over chunks answers fluently and unverifiably; a graph answers precisely but only in the terms someone defined. The pattern joins the two under an ontology a person owns: documents become passages, passages become typed facts that cite them, the facts are the record, projections serve queries, and an agent answers through fixed tools with claims that code checks before anything is shown.

The audience is a team building a knowledge store over its own documents, in any domain, who will own the ontology.

Non-goals

- A general chat assistant: the agent answers from the collection or abstains.
- A model as judge of truth: every check is code, with the guardrail as a second, bounded opinion.
- Free querying by the model: the tools are fixed.
- A document management system: sources are read, never managed.

Quality attributes

Quality attribute	Scenario	Response measure
Groundedness	A model proposes a claim whose quote is not in the passage	The claim is repaired once or removed; it never reaches the page
Scope safety	A public caller's model asks a tool for private content	The interceptor sets the scope; the tool returns public content only
Cost	A question runs away in tool calls	The hook cancels the seventeenth call; the agent answers from what it has
Rebuildability	A projection is lost	The next sweep rebuilds it from gold; the pointer moves when counts check
Governance	A term is proposed from a document	It stays a candidate until a person publishes a version containing it

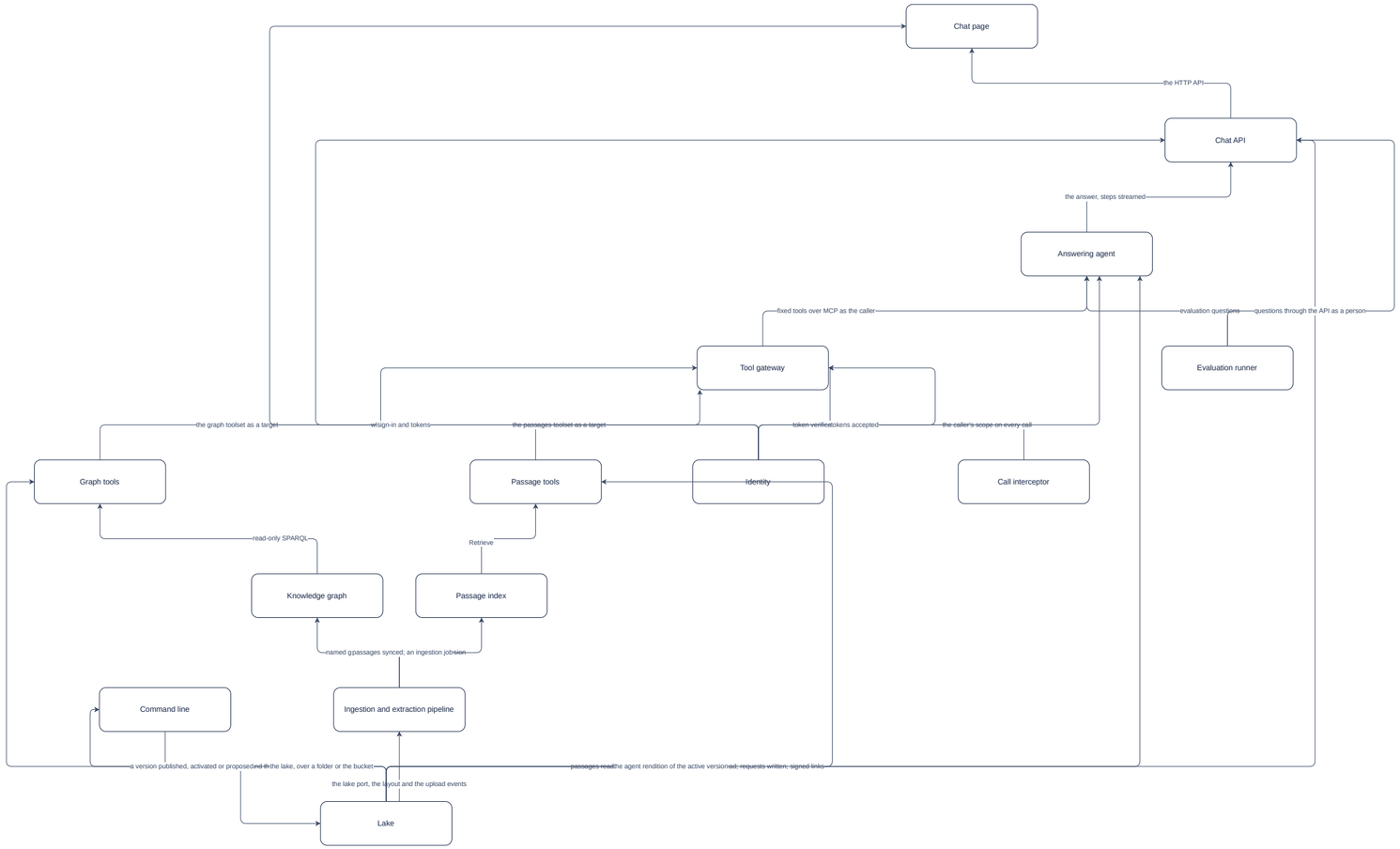
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Diagram



Generated from the Required Building Blocks and Interfaces tables with the model skill (`components.drawio` is the source); the as-built picture is RA-001's.

Required Building Blocks (1 of 2)

Building Block	Role in the architecture pattern
ABB-002 Lake	The record and its layers; the contract every container reads
ABB-024 Ingestion and extraction pipeline	Bytes to published facts in one idempotent sweep; composes four contexts today
ABB-011 Knowledge graph	The graph projection, rebuilt from the record
ABB-013 Passage index	Passages by meaning; the hit is a passage id
ABB-012 Graph tools	Fixed, read-only tools in the ontology's terms, scope-filtered
ABB-025 Passage tools	Search and read of passages, scope-filtered
ABB-026 Tool gateway	One MCP surface and one token over the tools

Required Building Blocks (2 of 2)

Building Block	Role in the architecture pattern
ABB-027 Call interceptor	The one scope decision, before any tool runs
ABB-014 Answering agent	Plans, proposes, checks, repairs; nothing shown unchecked
ABB-019 Chat API	The narrow door; quota, state, signed links
ABB-018 Chat page	Where grounding is visible
ABB-020 Identity	The person all the way down
ABB-028 Command line	The whole store on one machine; the release commands
ABB-021 Evaluation runner	Right, not only grounded

Every box is one C4 container under the calibration standard: the first pass's components (adapters, refinement, discovery, versioning, candidates, mapping, extraction, binding, projection, the checks, the valves, the workbench, the model client) are rows inside these blocks, and the deployment's infrastructure (triggers, build, module) is kept off the pattern.

Interfaces (1 of 2)

Interface	Provider	Consumer	Purpose
IF-01	ABB-002 Lake	ABB-024 Ingestion and extraction pipeline	the lake port, the layout and the upload events
IF-02	ABB-024 Ingestion and extraction pipeline	ABB-011 Knowledge graph	named graphs loaded per document per version
IF-03	ABB-024 Ingestion and extraction pipeline	ABB-013 Passage index	passages synced; an ingestion job
IF-04	ABB-002 Lake	ABB-012 Graph tools	the gold index, the snapshots and the active version
IF-05	ABB-011 Knowledge graph	ABB-012 Graph tools	read-only SPARQL
IF-06	ABB-013 Passage index	ABB-025 Passage tools	Retrieve
IF-07	ABB-002 Lake	ABB-025 Passage tools	passages read by id
IF-08	ABB-012 Graph tools	ABB-026 Tool gateway	the graph toolset as a target
IF-09	ABB-025 Passage tools	ABB-026 Tool gateway	the passages toolset as a target
IF-10	ABB-027 Call interceptor	ABB-026 Tool gateway	the caller's scope on every call
IF-11	ABB-026 Tool gateway	ABB-014 Answering agent	fixed tools over MCP as the caller
IF-12	ABB-020 Identity	ABB-026 Tool gateway	token verification

Interfaces (2 of 2)

Interface	Provider	Consumer	Purpose
IF-13	ABB-020 Identity	ABB-019 Chat API	who is asking
IF-14	ABB-020 Identity	ABB-018 Chat page	sign-in and tokens
IF-15	ABB-020 Identity	ABB-014 Answering agent	tokens accepted
IF-16	ABB-014 Answering agent	ABB-019 Chat API	the answer, steps streamed
IF-17	ABB-019 Chat API	ABB-018 Chat page	the HTTP API
IF-18	ABB-002 Lake	ABB-019 Chat API	served layers read; requests written; signed links
IF-19	ABB-002 Lake	ABB-014 Answering agent	the agent rendition of the active version
IF-20	ABB-002 Lake	ABB-028 Command line	the lake, over a folder or the bucket
IF-21	ABB-028 Command line	ABB-002 Lake	a version published, activated or proposed
IF-22	ABB-021 Evaluation runner	ABB-014 Answering agent	evaluation questions
IF-23	ABB-021 Evaluation runner	ABB-019 Chat API	questions through the API as a person

Controls

Control	Where	What it stops
The scope rule at every hop	ABB-020	A public caller reading private content
Claims checked by code	ABB-014	An invented quote, value or figure
Fixed tools	ABB-012	Writes, free queries, escape from scope
Valves	ABB-014	Runaway cost per question, per person, per deployment
Publish refuses an understated bump	ABB-024	A silent semantic change
Candidates stay out of the graph	ABB-024	Document text changing the ontology
Documents as data	ABB-024	Document text being executed or trusted

Decisions

ADR-001 to ADR-005 and ADR-010 are this architecture pattern's; ADR-006 to ADR-009 belong to the reference architecture, RA-001.